

Methods Annex — Measuring the Five Pillars and Composite Π

1. Scope and Purpose

1.1 This annex specifies how to: (a) compute scores for the Five Pillars; (b) aggregate them into the composite Π via geometric means; (c) quantify uncertainty; (d) run sensitivity and robustness checks; (e) detect gaming (Goodhart); (f) handle missing data; and (g) audit and reproduce results.

1.2 Design principles: transparency, monotone transforms, conservative anchors, and publication of uncertainty and sensitivity alongside point estimates.

2. Indicators (fixed list)

For each Pillar, the indicator bundle, units, and direction are fixed. All will be normalized to $[0,1]$.

2.1 Birth / Crew Formation

- **B1** Completed household formations per 1,000 adults ($\uparrow$)
- **B2** 3-year rolling Total Fertility Rate relative to charter target ($\uparrow$)
- **B3** First-child age distribution score (penalize extreme delay) ($\uparrow$)
- **B4** Dependency coverage ratio (childcare/eldercare slots per dependent) ($\uparrow$)

2.2 Language / Codec

- **L1** Intergenerational language retention (share of children proficient) ($\uparrow$)
- **L2** Work-lexicon fluency (share of adults passing civic/technical glossary test) ($\uparrow$)
- **L3** Translation gateway uptime for lingua-franca interfaces ($\uparrow$)
- **L4** Public proceedings linguistic accessibility (share with dual-language provision) ($\uparrow$)

2.3 Land / Habitat

- **H1** Preventive maintenance completion rate (scheduled/required) (↑)
- **H2** Unplanned outage hours per 1,000 users (↓)
- **H3** Habitat reserve coverage (days of critical spares/cooling) (↑)
- **H4** Environmental quality index (composite of air/water/thermal spill) (↑)

2.4 Skill / Reliability

- **S1** Critical guild density (certified maintainers per 10k pop) (↑)
- **S2** Mean time between failures (MTBF) in critical systems (↑)
- **S3** Shadow-ops drill pass rate (manual takeovers within SLA) (↑)
- **S4** Defect escape rate (post-release defects per 1k changes) (↓)

2.5 Security (Bounded)

- **Q1** Adjudicated offense rate per 10k (↓)
- **Q2** Tribunal interim-order SLA compliance ($\leq 72h$) (↑)
- **Q3** Green-lane uptime (corridor/SAR/medical) (↑)
- **Q4** Sanctions-before-shots ratio (sanctions resolved / kinetic incidents) (↑)

3. Data Sources (assumptions)

3.1 Administrative records: registries, court ledgers, port/corridor logs, SAR reports, CMMS maintenance systems, and guild rosters.

3.2 Telemetry aggregates: scheduler uptime, translation gateways, outage meters.

3.3 Survey modules (annual/biannual): language proficiency, household formation, lexicon tests.

3.4 Audit samples: randomized file reviews, drill observations, teardown lotteries (for lineage/provenance).

3.5 Privacy: personally identifying data are aggregated before use; a privacy budget and controlled randomization apply where needed (see §10).

4. Normalization and Construction

4.1 Each indicator (x) is mapped to a unitless score ($s \in [0,1]$) via min-max with winsorized anchors ($(A_{\text{low}}, A_{\text{high}})$).

- **For $\uparrow$ indicators:**

$$s = \text{clamp}(0,1, (x - A_{\text{low}}) / (A_{\text{high}} - A_{\text{low}}))$$

- **For $\downarrow$ indicators:**

$$s = \text{clamp}(0,1, (A_{\text{high}} - x) / (A_{\text{high}} - A_{\text{low}}))$$

4.2 Anchors are posted per indicator (e.g., S2 MTBF: ($A_{\text{low}}=200$) h, ($A_{\text{high}}=1500$) h), updated annually using a 3-year rolling window and expert oversight.

4.3 **Pillar scores** use geometric means with equal within-pillar weights by default:

$$\text{Pillar} = (\prod s_i)^{(1/k)} \text{ where } (k) \text{ is the number of sub-indicators.}$$

4.4 **Composite Π** is the geometric mean across the five Pillars:

$$\Pi = (\text{Birth} \times \text{Language} \times \text{Habitat} \times \text{Skill} \times \text{Security})^{(1/5)}.$$

4.5 **Rationale:** the geometric mean penalizes lopsided performance; no single pillar can mask collapse elsewhere.

5. Uncertainty Estimation

5.1 **Block bootstrap (indicator level):** For each locale-year, draw 1,000 resamples by sampling indicator rows with replacement **within pillars** (preserving within-pillar dependence). For administrative streams with near-census coverage, add $\pm 2\%$ mean-zero observational noise before resampling.

5.2 For each resample, recompute normalized (s_i), Pillars, and Π . Report **95% intervals** (2.5th/97.5th percentiles) for each Pillar and for Π .

5.3 Publish **signal-to-noise ratio (SNR)** per indicator: annual mean divided by bootstrap SD.

6. Sensitivity and Robustness

6.1 **One-at-a-time (OAT) anchor sensitivity:** For each indicator, shock (A_{low}) and (A_{high}) by $\pm 10\%$ and recompute Pillars and Π ; publish $\Delta\Pi$ and the top five drivers.

6.2 **Weight perturbations:**

- **Within-pillar:** random Dirichlet draws over sub-indicator weights.
- **Across pillars:** scenario weights (e.g., Security-weighted during crises). Publish the resulting bands.

6.3 **Leave-one-out:** Drop each sub-indicator in turn; report Π spread.

6.4 Present tornado plots **or** a sorted table of drivers of Π variance.

7. Goodhart and Gaming Detection

7.1 We define gaming signatures and counters per indicator; examples (≥ 5 shown):

- **B1 Household formations:** sudden spike without matching **B4** coverage $\rightarrow$ paper marriages.

Counter: require joint evidence (co-residence + shared financials) and check 12-month retention.

- **L2 Lexicon proficiency:** short-term cramming.

Counter: rotate test pools; include applied tasks; penalize sharp reversion next period.

- **H1/H2 Maintenance vs outages:** reported **H1** $\uparrow$ with flat **H2** $\rightarrow$ box-ticking.

Counter: cross-check with parts usage and vendor invoices; random site audits.

- **S3 Shadow-ops drills:** staged drills.

Counter: unannounced drills with external observers and cross-team tasking.

- **Q2/Q4 Tribunal & sanctions:** reclassifying incidents to improve Q4.

Counter: third-party log reconciliation; publish definitions and raw counts side-by-side.

7.2 **CUSUM/SPR sequential tests:** for each sub-indicator, run change-point detection; flag abrupt improvements lacking corroboration in paired metrics; route to audit.

8. Privacy and Data Minimization

8.1 Privacy budget $\leq$ **200 bits** of event data per person-year exposed to analytics after aggregation.

8.2 Salted, rotating pseudonyms for linkage; published deletion schedules.

8.3 Release only Pillar and Π aggregates with 95% intervals; no small-cell tables.

8.4 Audits run within on-site secure enclaves; only statistics exit.

9. Missing Data and Imputation

9.1 If any sub-indicator is missing:

- If pillar coverage $\geq 80\%$: impute missing (s_i) by last observation carried forward (max 2 periods) or the regional median if no history.
- If pillar coverage $< 80\%$: withhold the pillar and Π for that locale/period; flag **insufficient data**.

9.2 Mark all imputed components; rerun sensitivity excluding imputed points; publish deltas.

10. Worked Numeric Example (one locale, one year)

10.1 Raw → normalized (anchors and direction shown)

Normalization formula (once):

- $\uparrow: s = (x - A_{\text{low}}) / (A_{\text{high}} - A_{\text{low}})$
- $\downarrow: s = (A_{\text{high}} - x) / (A_{\text{high}} - A_{\text{low}})$

Clamp to [0,1].

Ind.	Value	A_low	A_high	Dir	s_i
B1	42 per 1,000	20	60	↑	0.550
B2	1.85 vs 2.10	1.40	2.40	↑	0.450
B3	0.68	0.40	0.90	↑	0.560
B4	0.78	0.50	0.95	↑	0.622
L1	0.74	0.40	0.95	↑	0.618
L2	0.62	0.30	0.90	↑	0.533
L3	0.985	0.950	0.995	↑	0.778
L4	0.81	0.50	1.00	↑	0.620
H1	0.88	0.60	0.98	↑	0.737
H2	14 h/1k	60	5	↓	0.164
H3	9.0 days	3	15	↑	0.500
H4	0.76	0.50	0.95	↑	0.578
S1	18 /10k	6	24	↑	0.667
S2	920 h MTBF	200	1500	↑	0.554
S3	0.72	0.40	0.95	↑	0.582
S4	0.84 /1k	2.0	0.2	↓	0.356
Q1	3.6 /10k	18	1	↓	0.153
Q2	0.91	0.70	0.98	↑	0.750
Q3	0.982	0.94	0.995	↑	0.764
Q4	14:1	3	20	↑	0.647

10.2 Pillar scores (geometric means)

- **Birth** = $(0.550 \times 0.450 \times 0.560 \times 0.622)^{(1/4)} = \mathbf{0.542}$
- **Language** = $(0.618 \times 0.533 \times 0.778 \times 0.620)^{(1/4)} = \mathbf{0.631}$
- **Habitat** = $(0.737 \times 0.164 \times 0.500 \times 0.578)^{(1/4)} = \mathbf{0.432}$

- **Skill** = $(0.667 \times 0.554 \times 0.582 \times 0.356)^{(1/4)} = \mathbf{0.526}$
- **Security** = $(0.153 \times 0.750 \times 0.764 \times 0.647)^{(1/4)} = \mathbf{0.488}$

10.3 Composite Π

$$\Pi = (\text{Birth} \times \text{Language} \times \text{Habitat} \times \text{Skill} \times \text{Security})^{(1/5)}$$

$$\Pi = (0.542 \times 0.631 \times 0.432 \times 0.526 \times 0.488)^{(1/5)} = 0.520 \text{ (to three decimals).}$$

10.4 Uncertainty (bootstrap sketch)

- Build 1,000 resamples at the **indicator** level with replacement; add $\pm 2\%$ noise to near-census streams.
- For each: re-normalize, recompute pillars and Π .
- Report 2.5th/97.5th percentiles for each pillar and Π (e.g., Π 95% CI = [0.497, 0.542] if obtained).
- Publish SNR per indicator.

10.5 Sensitivity snippet (OAT, anchors)

Perturb anchors and recompute Π ; report $\Delta\Pi$ (absolute change from 0.520).

Test	Anchor change	Π (new)	$\Delta\Pi$
S2 MTBF (Skill)	A_high +10% (1500→1650)	0.517	-0.003
Q3 Green-lane (Security)	A_low -10% (0.94→0.846)	0.524	+0.005

Interpretation: Π is modestly sensitive to Security corridor uptime anchors and less so to MTBF upper anchors in this scenario.

11. Falsifiers and Stop/Go Criteria

11.1 Falsifier A — Reliability: If two independent rail implementations cannot maintain a performance spread $< 5\%$ for three consecutive quarters, Π reporting pauses; re-open the standards market and retest.

11.2 Falsifier B — Justice: If **Q3** (green-lane uptime) < 0.95 for two consecutive periods, no charter reforms may claim success irrespective of Π .

11.3 Falsifier C — Goodhart: If ≥ 3 indicators show CUSUM anomalies uncorroborated by paired metrics (e.g., **H1** $\uparrow$ without **H2** $\downarrow$), flag the series; suspend from Π until audits resolve.

11.4 Stop/Go: If > 1 falsifier fires in the same period, publish Π with a striped watermark (“under audit”) and withhold any policy bonuses/rebates tied to Π .

12. Publication and Audit

12.1 Release: (a) Pillar and Π point estimates; (b) 95% intervals; (c) SNR table; (d) sensitivity summary (top five drivers); (e) exceptions log (missing/imputed components, audits pending).

12.2 Post the Methods Annex version number and **the exact anchors used** that period.

12.3 Provide a machine-readable bundle (JSON/CSV) containing: raw aggregates, normalized (s_i), pillar scores, Π , bootstrap intervals, and sensitivity outputs.

12.4 Reproducibility bounties: invite independent reruns; pay the first verified replication within tolerance.

13. Governance of Methods (versioning and change control)

13.1 Changes require: (1) red-team memo; (2) pilot back-test on three prior years; (3) 30-day public comment; (4) supermajority of the standards committee with conflict-of-interest disclosures.

13.2 Versioning:

- Structural method changes: **major** version.
- Anchor updates: **patch** version.

- Other method tweaks: **minor** version.

13.3 All historical releases archive the code, anchors, and outputs to enable exact reproduction.

14. Implementation Notes (operational)

14.1 **Posting anchors:** For each indicator, publish (A_{low}) and (A_{high}), last three years' values, and rationale (e.g., historical percentiles or policy targets).

14.2 **Winsorization:** Values beyond anchors are clamped at 0 or 1; never extrapolate above 1.

14.3 **Locale comparability:** Use the same anchors across comparable locales; if material heterogeneity exists, define families of locales with family-specific anchors and disclose.

14.4 **Crisis overrides:** Temporary pillar weights may be published for crisis windows (e.g., Security-weighted), but Π (**equal-weighted**) **remains the canonical headline**; crisis-weighted Π^* may be shown as a secondary series with clear labeling.

15. Quality Controls (checklist)

- Anchors posted and documented.
- Raw $\rightarrow$ normalized mapping reproducible from released bundle.
- Bootstrap CI computed (1,000 draws) and SNR published.
- OAT, weight, and leave-one-out sensitivity completed; top drivers listed.
- CUSUM/SPR run; anomalies triaged; audits logged.
- Missing data rules applied and flagged; imputation deltas published.
- Falsifiers evaluated; stop/go actions documented if triggered.
- Version numbers incremented and archived.

16. Plain-Language Summary (2–3 sentences)

Π is a single reliability score built from five Pillars—Birth, Language, Habitat, Skill, and Security—each measured by a small set of indicators. We use geometric means so weakness in one area cannot be hidden by strength in another. Every release includes uncertainty bands and sensitivity tests so readers can see how stable the score is and how it would move under reasonable assumptions.